Assignment 1

Variance and Bias (Diagram, overfit, underfit) - For best fit model should we have low bias or high variance, low bias or low variance, high bias or high variance, low bias or high variance

Introduction

In machine learning and statistical modeling, understanding bias and variance is essential for building models that generalize well to new data. These two sources of error determine whether a model is too simple, too complex, or appropriately balanced. The interplay between bias and variance is commonly known as the bias–variance tradeoff.

When training predictive models, the goal is not merely to fit the training data but to perform accurately on unseen data. Models that memorize training data often fail in real-world scenarios, while overly simple models miss important patterns. Therefore, achieving the right balance between bias and variance is critical.

Bias

Definition

Bias refers to the error introduced by approximating a real-world problem with a simplified model. It measures how far the model's predictions are from the true underlying relationship.

Formally:

Bias = Error due to incorrect assumptions in the learning algorithm

Characteristics of High Bias

A model with high bias:

- Is overly simple
- Makes strong assumptions about the data
- Fails to capture important patterns
- Performs poorly on both training and test data

This situation leads to underfitting.

Example

Suppose the true relationship between variables is quadratic, but we fit a straight line. The model cannot represent the curvature, resulting in high bias.

Understanding Variance

Definition

Variance refers to how much the model's predictions change when trained on different datasets. It measures the model's sensitivity to fluctuations in the training data.

Formally:

Variance = Error due to model sensitivity to small changes in training data

Characteristics of High Variance

A model with high variance:

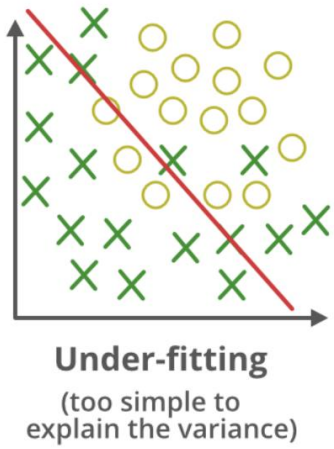
- Is overly complex
- Fits noise in the training data
- Performs very well on training data
- Performs poorly on test data

This situation leads to **overfitting**.

Example

A very deep decision tree may perfectly memorize the training set but fail on new data because it learned noise rather than the true pattern.

Underfitting (High Bias Problem)



What is Underfitting?

Underfitting occurs when a model is too simple to capture the underlying structure of the data.

Key property: High bias, low variance.

Characteristics

- Training error: High
- Test error: High
- Model complexity: Too low
- Pattern learning: Poor

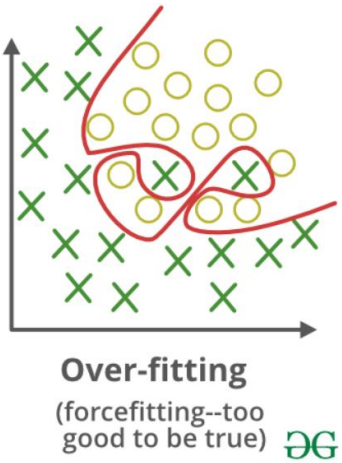
4.4 Causes

- Model too simple
- Too few features
- Excessive regularization
- Insufficient training time

4.5 Remedies

- Increase model complexity
- Add relevant features
- Reduce regularization
- Use nonlinear models

Overfitting (High Variance Problem)



What is Overfitting?

Overfitting occurs when a model learns not only the true pattern but also the noise in the training data.

Key property: Low bias, high variance.

Data points: closely matched by model

Characteristics

- Training error: Very low
- Test error: High
- Model complexity: Too high

- Generalization: Poor

Causes

- Model too complex
- Too many parameters
- Small training dataset
- Insufficient regularization

Remedies

- Simplify the model
- Add more training data
- Apply regularization (L1/L2)
- Use cross-validation
- Prune decision trees
- Early stopping

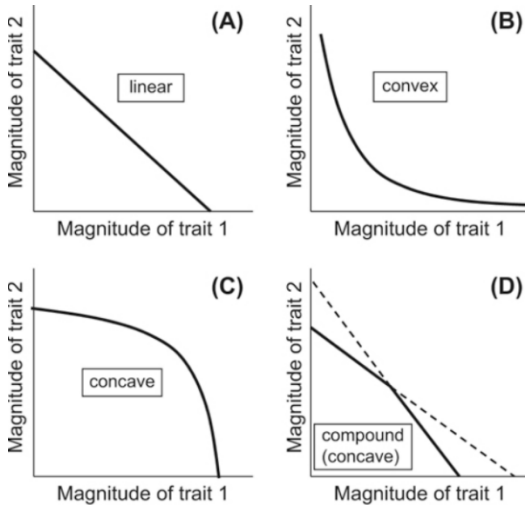
The Bias–Variance Tradeoff

Core Idea

Bias and variance typically move in opposite directions:

- **Increasing model complexity → ↓ Bias but ↑ Variance**
- **Decreasing model complexity → ↑ Bias but ↓ Variance**

Tradeoff Curve



Practical Examples

Linear Regression

- Simple linear regression → high bias

- Polynomial regression (high degree) → high variance
- Moderate polynomial → balanced model

Decision Trees

- Shallow tree → underfitting (high bias)
- Very deep tree → overfitting (high variance)
- Pruned tree → good balance

Neural Networks

- Too few neurons → high bias
- Too many layers without regularization → high variance
- Proper architecture + dropout → balanced

Techniques to Control Bias and Variance

To Reduce High Bias

- Increase model complexity
- Add more features
- Reduce regularization
- Use nonlinear models
- Train longer

To Reduce High Variance

- Collect more data
- Apply regularization
- Use dropout
- Feature selection
- Ensemble methods (bagging, random forests)
- Cross-validation

Role of Cross-Validation

Cross-validation helps estimate how the model will perform on unseen data.

Benefits:

- Detects overfitting
- Helps select model complexity

- Provides reliable performance estimate

Common methods:

- k-fold cross-validation
- stratified k-fold
- leave-one-out

Real-World Intuition

Think of bias and variance like archery:

- **High bias:** arrows consistently miss the bullseye in the same direction (systematic error)
- **High variance:** arrows scattered everywhere (inconsistent)
- **Low bias & low variance:** arrows tightly clustered at the center (ideal model)

Conclusion

Bias and variance are fundamental concepts that determine the success of machine learning models.

High bias leads to underfitting, where the model is too simple to learn meaningful patterns.

High variance leads to overfitting, where the model memorizes noise and fails to generalize.

The key insight is the **bias–variance tradeoff**: improving one often worsens the other. Therefore, the objective is not to minimize bias or variance independently but to find the optimal balance that minimizes total prediction error.

Final takeaway:

The best-fit model is achieved when both bias and variance are low and balanced.

Achieving this balance requires careful model selection, proper regularization, sufficient data, and rigorous validation techniques.

Mastery of the bias–variance tradeoff is essential for building robust, real-world machine learning systems.